

1. THE VERIFICATION CONTRACT

The program is a verifier of finite mathematical statements, not an oracle for Hilbert’s Tenth Problem over $\mathbb{Q}$. A positive row is accepted only after its stated hypotheses and exact identities have been replayed. A bounded scan licenses a statement about the scanned rows and no statement about the unscanned complement. In particular, the canonical grid is closed instance-by-instance, while the uniform all- w assertion remains an analytic frontier; the audited simultaneous prime-value input has degree 8 and is still open.

1.1. The label calculus. The ledger uses three labels as a small calculus of admissible inference.

PROVED means that an exact implication, identity, certificate, or finite exhaustion was replayed by the machine. For example, the four off-grid rows at $w \in \{101, 103, 107, 109\}$, $u = -1$, are PROVED per row after independent replay; this is a finite horizon statement, not a uniform theorem in w .

CONDITIONAL means that every remaining hypothesis is stated and named. The six-unknown conclusion is conditional on the explicitly isolated global assembly lemma, so no quantifier record is claimed. At the analytic end of the class reduction, the simultaneous primality of $q_1(t)$ and the degree-8 polynomial $F_{\text{cell}}(t)$ awaits Schinzel’s Hypothesis H; the finite local audit does not supply that conjectural prime-value conclusion.

EVIDENCE means a measured statistic with its sampling frame attached. The matched obstruction artifact reports marginal ratio 0.9993, a union residual of $-2.9/8760$, and a marginal-preserving permutation statistic $z = -0.09$ with empirical tails 0.61/0.53. These numbers validate the matched finite model as EVIDENCE; they do not prove independence or a density theorem.

Labels only move upward through a valid proof. In particular, **PrimalityBound**, **FactorBudget**, a timeout guard, a zero-Jacobi outcome, or a missing producer cannot upgrade EVIDENCE to PROVED and cannot turn an open case into a negative result. A refusal contributes *unknown*; only a Fermat witness, a proper factor, or another exact certificate contributes a mathematical verdict.

2. PROVEN-PRIMALITY ARITHMETIC

2.1. The deterministic range. The entry point `_is_prime` first applies strong Miller–Rabin at the thirteen bases from 2 through 41 [2]. Its strict deterministic cutoff is

$$\text{_MR.LIMIT} = 3317044064679887385961981.$$

For n below this bound, passing all bases is a proved primality verdict. At or above the bound, passing is only a screen and the code proceeds to a certificate; it never accepts “probable prime” as prime.

This distinction is load-bearing because factorization feeds exact Hilbert symbols and ramification sets. The regression suite therefore includes the least strong pseudoprime at the cutoff, a prime above the cutoff that must be Pocklington-certified, hard cases that must refuse, and a balanced semiprime that must exhaust the factor budget rather than be guessed.

2.2. Pocklington and the relaxed threshold. For a screened n , the engine writes

$$n - 1 = FR,$$

where every prime dividing the accumulated part F is recursively certified. For each such prime q , it seeks a base a satisfying

$$a^{n-1} \equiv 1 \pmod{n}, \quad \gcd(a^{(n-1)/q} - 1, n) = 1.$$

The generalized Pocklington conclusion is immediate when $F^2 > n$. A failed Fermat congruence or a proper gcd proves compositeness; failure to find a usable witness is a **PrimalityBound** refusal.

TABLE 1. Frozen finite authorities and their intended role.

Authority	Rows	Role
<code>L6_WITNESSES</code>	61	bounded canonical tied evidence
<code>L9_STEERED</code>	345	steered cell witnesses, fully replayed
<code>L11_CLASSES</code>	190	reachable aligned prime classes
<code>L12_ESCAPE</code>	103	non-coprime aligned escape classes
<code>L13_ESCAPE2</code>	60	fully soluble walled-cell witnesses
<code>L13_RESIDUAL</code>	6	legacy pointwise audit slice
<code>L13_ZERO_BAD</code>	16	frozen ladder regression rows
<code>L13_H_CLASSES</code>	293	one certified member per aligned grid class

Before attempting harder splitting, the implementation trial-divides $n - 1$ by every prime below 10^6 . It computes the integer ceiling of $n^{1/3}$ exactly and stops that trial loop as soon as the accumulated certified part reaches the cube-root threshold. Uncertifiable recursive chunks and factor-budget-exhausted chunks are excluded from F , never counted as if they had factored.

At the weaker threshold $F^3 \geq n$, the Brillhart–Lehmer–Selfridge relaxation, in the form recorded as Crandall–Pomerance Theorem 4.1.5, applies. The code first checks the required coprimality of F and R , writes $R = c_2F + c_1$, and tests

$$D = c_1^2 - 4c_2.$$

Under the certified congruence conditions a composite n can then have exactly two factors of the form $uF + 1$ and $vF + 1$. A nonnegative square D with the correct parity reconstructs those two factors and proves compositeness; otherwise the discriminant test proves primality. This is a certificate theorem, not a heuristic strengthening of Miller–Rabin.

2.3. Explicit refusal types. `PrimalityBound` means that the screen passed outside its deterministic range but the certified part or witnesses were insufficient. `FactorBudget` means Brent’s rho routine exhausted its bounded work on a large remaining composite. Callers either propagate these exceptions or mark the candidate as refused. The suite contains adversarial paths specifically to ensure that a partial factorization is not returned as complete and that a refusal is not counted as a soluble member.

3. SUITE ARCHITECTURE AND FROZEN AUTHORITY

A default invocation is `python3 h10q.py`; the extended invocation adds `--extended`. Both begin with arithmetic self-tests and the ledger verification, then replay the successive finite layers. The current engine note records approximately 51 seconds for the default suite and approximately 118 seconds for the extended suite on the recorded run.

The difference is scope, not soundness. The default L6 replay covers the odd prime targets below 100, while the extended replay reaches the full 61-row authority below 300. L13 class sampling uses 12 escape rows by default and 40 in the extended run; fully soluble escape witnesses use 24 versus all 60. Most importantly, the default L14 check takes a fixed seeded sample of 60 frozen H-class members, while the extended check replays all 293.

The first two authorities have a stronger serialization contract. Each suite run derives canonical JSONL text from the in-code objects and requires `data/16_witnesses.jsonl` and `data/19_steered.jsonl` to match byte for byte. Drift, absence, or an underived field fails the run. The only supported export command is

```
python3 h10q.py --export-evidence
```

which writes those canonical serializations; it does not canonize arbitrary search logs.

4. ARTIFACT SCHEMAS AND LICENSES

A schema is part of the proof boundary. Row types say whether a record is configuration, a replayable certificate, a failed attempt, or an aggregate. The principal persisted objects are summarized below; “licenses” always means “licenses at the recorded finite scope.”

4.1. Reading rows without overclaiming. Metadata is a scope declaration, not a certificate by itself. A **meta** row identifies the authority, coordinate convention, engine, window, and generator assumptions. The claim-bearing payload is then found in a class, canonical, kernel-replay, or closure record, while a summary is a deterministic reduction of those payload rows. Consequently a reader should not cite a global total without retaining the class-level artifact from which it was reduced, and should not cite an attempted class as a certified class.

The canonical closure document makes this lineage especially explicit. Its record stores the cell and family, the aligned progression $Q = q_1 + k_{\text{zero}}N$, the resulting member parameters, the ladder rung, and the source scan. The Python authority stores the corresponding tuple needed for replay. During L14 verification the suite reconstructs Q , proves it prime, forms $b = \varepsilon fQ$, re-runs the smooth emergent test and cofactor ladder, and accepts only the verdict **zero**. The optional full tied and ramification calculations are cross-checks rather than hidden prerequisites for this accepted certificate.

Large integers and rational parameters are serialized as decimal strings or explicit numerator–denominator data where the producing schema requires them. Status words such as **bad**, **deferred**, **nonprime**, and **refused:FactorBudget** are intentionally not coerced to booleans. This preserves the distinction between a proved negative result, a skipped member, and an arithmetic refusal. Schema compatibility therefore includes both field names and status semantics; a consumer that collapses those states has changed the estimand even if its JSON parser succeeds.

The horizon cross- k file is deliberately different. Its meta row freezes the probe space and determinism contract; untagged rows record primality, tied, ramified, and size outcomes. For H109 it contains 1,275 probes through $k \leq 50$, but no row with both requested auxiliary successes. That is an auxiliary finite non-hit, not a defect in the accepted zero-bad closure.

5. EXACT REPRODUCTION PROTOCOL

Run from the repository directory **math/h10q**. The downstream matched chain is:

```
python3 117_ratemodel_padic.py
python3 117_badroots_closures.py
python3 117_matched.py
```

The middle command imports the checked-in p-adic module and rebuilds closure- authority roots from the persisted sieve; the final command consumes those roots and the same persisted member rows. Thus the first arrow is also a code dependency. It does not claim recovery of the original session-side producer for the older censored-rate artifact.

The independent class-risk and Schinzel audits are regenerated by

```
python3 117_classrisk.py
python3 117_schinzel_audit.py
```

The former writes **data/117_classfactors.jsonl** and **/tmp/117_classrisk.report**; the latter writes **data/117_schinzel_audit.jsonl** and its report under **/tmp**.

The established off-grid horizon is replayed one cell at a time:

```
python3 117_horizon101.py
python3 117_horizon103.py
python3 117_horizon107.py
python3 117_horizon109.py
python3 117_horizon109_crossk.py
```

Each closure generator overwrites its named data artifact and emits or writes a report; the cross- k generator emits no elapsed or host-dependent fields. The ledger records about 17 minutes for that 1,275-probe auxiliary scan and records byte-identical regeneration for it. No runtime is supplied here for commands for which the ledger records none.

6. REVIEW-DRIVEN CORRECTION AS METHODOLOGY

The L17 cycle treated review objections as attacks on the estimand, not as requests for cosmetic caveats. Five corrections illustrate the method.

Recurring-prime-only support. The first small-prime product used only primes recurring in the observed sample. Omitting nonrecurring bad primes mechanically inflated the predicted clean mass; it was not evidence of cross-prime correlation. The fix was to construct closure-authority root tables across every prime in the declared window and to target larger primes actually observed in the same cell, with simple roots assigned exact mass $1/(p+1)$ and singular roots recursively resolved or intervalized.

Wrong-sign conditioning. A proposed explanation attributed the clean-rate offset to conditioning on prime Q , but the sign was backwards. At the Q -root residue, $2ab \equiv 0$, so the implementation marks the residue safe rather than bad. The corrected interpretation is a difference between Haar measure on $\mathbb{Z}_p$ and the finite k -window. Its measured size is approximately $+0.0195$, or 3.63 standard deviations in the artifact; it is characterized, not promoted to a dependence claim.

Cohort versus closure progression. The first matcher combined sieve members rebuilt from canonical closure classes with roots computed on an earlier wave cohort. In the documented ESC example at cell $(3, (-3, 1))$, the cohort used $q_1 = 41$, $\varepsilon = +1$, whereas the closure used $q_1 = 59$, $\varepsilon = -1$. The fix was to rebuild every root in the closure record's own (ε, f, q_1, N) coordinate. The repaired table reconciles 5,375/5,375 observed odd-negative occurrences.

Discarded resolved mass. An excluded singular root can have a rigorously resolved prefix even when its remaining Taylor tree is unfinished. An intermediate matcher discarded that known mass along with the unresolved remainder. The corrected artifact stores both pieces and forms the interval

$$\left[\prod_p (1 - k_p - r_p), \prod_p (1 - k_p) \right].$$

Across the 67 excluded pairs the maximum unresolved residual is 6.9×10^{-15} , so the finite-window interval is valid without pretending to control any larger prime.

The vacuous integer-support p-value. A clipped two-sided empirical value on discrete integer support reported 1.0, which carried little diagnostic information. The final report keeps the marginal-preserving null but presents its standardized statistic and both empirical tails. Together with the repaired support and masses, the final matched values are marginal ratio 0.9993, union residual $-2.9/8760$, $z = -0.09$, and tails 0.61/0.53. These are the final EVIDENCE numbers; the abandoned instruments license nothing.

7. LIMITATIONS

- (1) Five older L17 files are persisted evidence without checked-in original producers: `l17_sieve.jsonl`, the cohort `l17_badroots.jsonl`, `l17_ratemodel_censored.jsonl`, `l17_cofactor.jsonl`, and `l17_cofactor_nonclosure.jsonl`. Downstream scripts consume reviewed copies, but the original session-side acquisition scripts were not recovered.
- (2) The class factors and matched Haar comparison are statistics only for $p \leq 10^4$. They provide no bound whatsoever for primes above 10^4 , so they cannot by themselves prove a positive infinite product or a uniform density.
- (3) The nonclosure cofactor distribution is measurement-limited. Only 11/400 sampled rows were exactly resolved and 389 were prover refusals; the predeclared power gate failed. The

distributional conclusion is therefore INCONCLUSIVE, even though parity remained consistent on resolved rows.

- (4) Some closure records contain auxiliary tied-status or ramification refusals. For example, the H109 closure records budget/refusal values in those fields while its proved zero-bad ladder verdict remains the accepted closure form. Such fields must not be silently rewritten as success, and their refusals are not supporting evidence.
- (5) The Schinzel audit proves finite local admissibility and irreducibility for the canonical pairs, not simultaneous prime values of the degree-8 polynomials. The record claim remains CONDITIONAL, and the finite off-grid horizon does not imply the all- w hypothesis.

The reproducibility claim is therefore intentionally narrow but useful: checked-in authorities can be replayed, specified downstream artifacts can be regenerated, and every refusal remains visible. The paper's mathematical claims are exactly those finite certificates labeled PROVED; its statistics remain EVIDENCE; and its analytic extrapolations remain CONDITIONAL.

REFERENCES

- [1] R. Crandall and C. Pomerance, *Prime Numbers: A Computational Perspective*, 2nd ed., Springer, New York, 2005.
- [2] G. Jaeschke, *On strong pseudoprimes to several bases*, *Mathematics of Computation* **61** (1993), no. 204, 915–926.

TABLE 2. Principal artifact schemas.

Artifact	Row types and key fields	What the row licenses
117_sieve.jsonl	meta ; class-summary with cell , family , members , forbidden residues, rates, alarms; global-summary	Per-class residue determination and the recorded member window; aggregates are measured summaries, not tail bounds.
117_badroots_closures.jsonl	meta ; class with progression parameters, root lists, exact or residual masses, exclusions; summary with reconciliation	Certified bad-signed $k \bmod p$ roots and their stated p -adic masses for the closure progression; excluded roots license only an interval.
117_matched.jsonl	meta , meta-methods , global , class ; keys include matched support, expectations, residuals, window factors, permutation null, and partial flags	EVIDENCE for fit on the identical member sample. It licenses neither cross-prime independence nor behavior above the recorded prime window.
117_classfactors.jsonl	Untagged per-class rows: cell , progression parameters, partial , factor_lower , factor_upper , factor_window , leading bad primes	Exact bookkeeping for each finite window interval; closure-effort correlation is descriptive EVIDENCE only.
117_schinzel_audit.jsonl	meta ; canonical with the two polynomials, coefficients, irreducibility certificate, value gcds, local counts; residual with pointwise witness and schinzel_applicable=false	Finite irreducibility, fixed-divisor, pair-gcd, and local-condition checks. Residual rows are pointwise only; no simultaneous prime values follow.
117_horizon*.jsonl	Usually meta , class-attempt , aligned-class , member , kernel-replay , an untagged closure row, and summary ; the compact H107 file has only meta and closure	Only a verified replay plus the exact closure row licenses an off-grid closure. Attempt and refusal rows are provenance, not negative evidence.
113h_all_closures.json	Top-level n_cells and records ; each record has family , cell , a , eps , f , q1 , N , k_zero , Q , rung , tied , ramified_empty , source	The canonical closure authority. Its thirteen-field record licenses the zero-bad ladder closure; auxiliary tied or ramified fields may themselves be refused without weakening that license.